TITANIC DATA EXPLORATION AND PREPROCESSING REPORT

ABSTRACT

This project focuses on performing data exploration and preprocessing on the Titanic Dataset using Python. The dataset was analyzed to understand its structure, identify missing values, convert data types, visualize target variable distribution, and prepare the dataset for future machine learning tasks. The project demonstrates the importance of preprocessing techniques in improving data quality before predictive modeling and analysis.

PROJECT OBJECTIVES

• Explore the dataset structure including rows, columns, and data types.
• Detect and handle missing values using appropriate statistical methods.
• Perform data type conversion for categorical variables.
• Analyze the target variable distribution and class imbalance.
• Create visualizations using Matplotlib and Seaborn.
• Save the cleaned dataset for future analysis and machine learning applications.

TOOLS AND TECHNOLOGIES USED

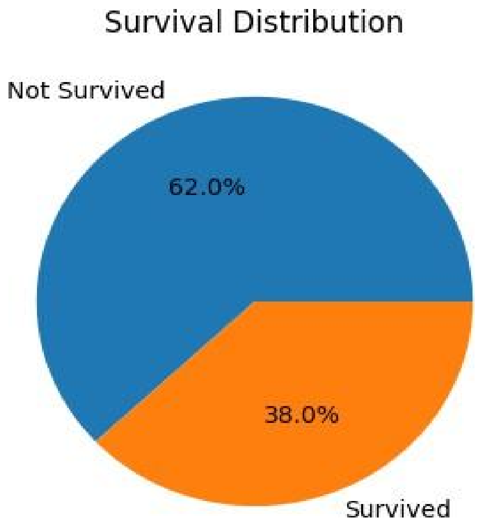
Python, Pandas, NumPy, Matplotlib, and Seaborn were used for performing data preprocessing, visualization, and analysis tasks in this project.

PROJECT WORKFLOW

Phase	Tasks Performed
Dataset Exploration	Displayed dataset shape, rows, columns, and data types
Missing Value Handling	Filled missing values in Age, Embarked, and Cabin columns
Visualization	Generated heatmaps, count plots, and pie charts
Data Type Conversion	Converted Survived and Pclass columns into categorical types
Dataset Export	Saved cleaned dataset as a new CSV file

TARGET VARIABLE ANALYSIS

The target variable **'Survived'** was analyzed to identify the class distribution. Count plots and pie charts were generated to visualize the percentage of survived and non-survived passengers. This helped identify class imbalance in the dataset and provided better insight into passenger survival patterns.



PROJECT METHODOLOGY

Step 1: Imported essential Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

Step 2: Loaded the Titanic dataset using the *read_csv()* function.

Step 3: Explored the dataset by checking rows, columns, data types, and missing values.

Step 4: Handled missing values using mean, mode, and replacement techniques.

Step 5: Visualized missing values and target variable distribution using heatmaps and plots.

Step 6: Converted important columns into categorical data types for better analysis.

Step 7: Saved the cleaned dataset as a CSV file for future predictive modeling.

CONCLUSION

The Titanic dataset was successfully explored and preprocessed using Python. Missing values were handled efficiently, visualizations were generated for better understanding, and the dataset was cleaned for future machine learning applications. This project highlights the significance of preprocessing and exploratory data analysis in the data science workflow.